

# Consensus-Based Distributed Control for Overvoltage Mitigation in LV Microgrids

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**Abstract**— In this paper, a consensus-based distributed control is proposed to mitigate overvoltage at the point of connection of PV systems in a low-voltage microgrid. The proposed control scheme uses a hierarchical control comprising the local control at each of PV inverters acting as the lower level control and the consensus-based distributed control acting as the higher level control. To cope with overvoltage problems due to the increasing PV penetration, the lower level control includes both the  $Q$ - $V$  and  $P$ - $V$  droop functions. By using a consensus-based algorithm on sparse communication infrastructure, the higher level control layer is activated to allocate control resources optimally among PV inverters, i.e. active power curtailment of PV inverters being equal and as minimum as possible. The feasibility of the proposed control scheme is successfully verified through simulations on a French distribution network using MATLAB/Simulink.

**Index Terms**—Microgrid, Photovoltaic (PV), Overvoltage problem, Consensus algorithm, Distributed control.

## I. INTRODUCTION

Recently, the distribution network has been shifting to a more actively controlled system, stemming from the information communication technology and distributed energy resources being increasingly integrated. Microgrid (MG) as a small scale distribution network with the capability of hosting and effectively controlling Distributed Generators (DG), Energy Storage Systems and flexible loads is considered as a promising solution to facilitate this shifting of the distribution network. One of the most popular types of DG is photovoltaic (PV). Due to the continuous reduction in PV panel cost, the PV installations in LV MG is rapidly growing. If the total installed PV capacity exceeds the hosting capacity of the feeder, i.e., the maximum power of PV that the feeder is capable to accommodate, the voltage quality can be degraded [1]. Particularly, when the power generation of the feeder is higher than its power demand, reverse power flow occurs, which consequently can result in overvoltage [1]. Without the proper voltage regulation methods, the overvoltage problem leads to the tripping of PV and other equipment for security reasons [2]. With increasing integration of PV, the tripping can become more frequent, causing the significant loss of the power

generation of PV. Thus, overvoltage becomes an impediment for the maximum output power of PV and reduces the hosting capacity of the feeder. This important problem is studied in this paper. Solutions for this problem can be categorized into grid-based methods and PV inverter-based methods. The former is regarded as the classical method, which consists of network reinforcement and the transformer tap changer adjustment. These two approaches are either expensive, time-consuming or less practical to be implemented [2]. The latter involves controlling the PV inverters to regulate its output power. This approach has been attracting the researchers and companies because it maximizes the use of existing equipment, i.e., the PV inverters, which can allow it to be a flexible and inexpensive approach. Being inspired by this fact, this paper also focuses on PV inverter-based methods to solve the overvoltage problem due to the high PV penetration in a LV MG.

In [2], the curtailment of PV output active power using  $P$ - $V$  droop control is proposed to mitigate the overvoltage problem. Although this method can be easily implemented and inexpensive, it provokes the loss of revenue of PV owners from selling their surplus power and the unequal power curtailments among the PV systems. Because the voltage rises along the radial feeder, PV connected to the end of the feeder is likely to bear a higher share of curtailment. Meanwhile, PV connected closer to the transformer may no need to curtail any power. Few PV owners, therefore, are placed at disadvantageous positions. Authors in [3] introduce a supplementary sensitivity matrix-based method to fairly curtail PV output active power for overvoltage mitigation. This method, however, requires knowledge of the line impedance and utilizes a centralized control architecture. The fair power curtailment algorithm needs to receive measured data from all PV and send back the setpoints to all PV. It, thus, highly relies on an all-to-all communication network and a central controller, which makes the system development expensive. Furthermore, the amount of total curtailed power derived by this method is considerably higher than the amount obtained by the conventional  $P$ - $V$  droop control. In [4] the control of reactive power absorption of the PV inverters using  $Q$ - $V$  droop control is studied. Since the LV distribution network is characterized by a high ratio of  $R/X$ ,

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only reactive power absorption can be inefficient to alleviate the overvoltage problem. In [5], both  $P$ - $V$  and  $Q$ - $V$  droop control are implemented to solve the overvoltage problem. This combination of  $P$ - $V$  and  $Q$ - $V$  droop control emerges as an effective solution because it can alleviate the overvoltage with minimum curtailed active power. However, this method again uses a centralized control architecture. The reliability and control performance then can be greatly degraded if any failure comes from the communication network or the central controller. The distributed control, on the other hand, has attracted notable research interests in order to mitigate the bottlenecks of centralized control and obtain a higher reliability [6]. With a widespread integration of Renewable Energy Sources (RES), the centralized control is potentially impractical and can be replaced by a distributed control [7]. The distributed control uses a sparse communication network with significantly lower bandwidth requirements for data exchange between the units, resulting in the lower cost and less complexity of the communication infrastructure. In such control architecture, each node exchanges its locally measured data only to adjacent nodes [8], i.e., no central controller nor all-to-all communication network is needed. If a new RES is added in a LV MG, only the commissioning of communication links between itself and the adjacent RES is required while the communication protocols applied in the system need no adjustments [8]. In contrast, for the centralized control, there must have a complicated adjustment of the central controller. Thus, the plug-and-play framework, which is a desired feature of the future distribution network, can be easily realized using the distributed control. Review of the control techniques and the applications of the distributed control is presented in [6] and [9]. The distributed techniques can be classified into four categories: consensus-based techniques, distributed model predictive control-based techniques, agent-based techniques and decomposition-based techniques [6]. In [10]–[13], the distributed control using the consensus algorithm is adopted to address various issues in MG, such as accurate power-sharing, optimal network operation, and voltage regulation. In contrast to the approach of this paper, the voltage regulation in [11] and [12] aims at minimizing the voltage deviation from average. Meanwhile [13] focuses on solving the overvoltage problem by coordinating PV and battery systems.

Being motivated from the aforementioned literature reviews, this paper proposes a consensus-based distributed control for the overvoltage mitigation using the PV inverters. The study objective is to mitigate the overvoltage problem with the active power curtailment of PV inverters being equal and as minimum as possible. Compared with the literature, the contribution of this paper is that it introduces a hierarchical control including a local control at each of PV inverters and the consensus-based distributed control to realize the coordinative control for overvoltage mitigation. The proposed local control, acting as the lower level control, includes both  $Q$ - $V$  droop control and  $P$ - $V$  droop control in which  $Q$ - $V$  droop control is activated prior to  $P$ - $V$  droop control. The consensus-based distributed control, acting as the higher level control, is developed using the consensus algorithm along with a sparse communication at the network feeder level. The proposed control scheme involves the consensus-based distributed control recalculating the  $P$ - $V$  droop parameters to pursue the

control objectives. Results obtained by the proposed control scheme are compared with the corresponding derived by other control schemes, highlighting the feasibility of the proposed control scheme.

The rest of the paper is structured as follows: Section II briefly introduces the background while Section III describes in details the proposed control scheme. Section IV illustrates the test network and data while Section V discusses the main results. Section VI draws the main conclusions.

## II. BACKGROUND

### A. $P$ - $V$ Droop Control

The  $P$ - $V$  droop control regulates the output active power of a PV inverter,  $P_{net}$  in response to the voltage level measured at the point of connection (POC),  $V_m$ . As shown in (1), during the normal operation, the PV inverter operates at the maximum power point,  $P_{MPP}$  defined by the maximum power point tracking algorithm of the PV inverter. When  $V_m$  exceeds the threshold level of  $V_{thP}$ , the PV inverter reduces  $P_{net}$  as a linear function of  $V_m$ .

$$P_{net} = \begin{cases} P_{MPP} & \text{if } V_{min} < V_m \leq V_{thP} \\ P_{MPP} - m \times (V_m - V_{thP}) & \text{if } V_{thP} < V_m < V_{max} \\ 0 & \text{if } V_m \geq V_{max} \end{cases} \quad (1)$$

where  $[V_{min} \ V_{max}] = [0.9 \ 1.1]$  is the voltage acceptable range according to the standard EN 50160 and  $m$  is the droop slope coefficient.

### B. $Q$ - $V$ Droop Control

The  $Q$ - $V$  droop control regulates the output reactive power,  $Q_{net}$  of a PV inverter as a function of  $V_m$ . As expressed in (2),  $Q_{net}$  is set at zero as  $V_m$  varies between the injection threshold level,  $V_{thQ}$ , and the absorption threshold level,  $V_{athQ}$ . If  $V_m$  across  $V_{thQ}$  or  $V_{athQ}$ ,  $Q_{net}$  is regulated following a linear function.

$$Q_{net} = \begin{cases} Q_{max} \times \frac{V_m - V_{athQ}}{V_{max} - V_{athQ}} & \text{if } V_{athQ} < V_m \leq V_{max} \\ 0 & \text{if } V_{thQ} \leq V_m \leq V_{athQ} \\ Q_{max} \times \frac{V_{thQ} - V_m}{V_{thQ} - V_{min}} & \text{if } V_{min} \leq V_m < V_{thQ} \end{cases} \quad (2)$$

where  $Q_{max}$  denotes the maximum reactive power of PV inverters and is restricted by the inverter's apparent power,  $S$  and active power generating,  $P_{net}$  at a given irradiance. Furthermore,  $Q_{max}$  is limited following the power factor (PF) requirements stipulated in the European Standard EN 50438. It requires the distributed generation units connecting to the LV network to operate with PF ranging from 0.9 lagging to 0.9 leading [14]. This PF requirement is integrated into this work.

### C. Consensus-based Distributed Control

In a consensus algorithm, each node communicates and shares its local measurement as the state variable with only adjacent nodes [9]. Consensus algorithm aims at converging various nodes to a common equilibrium point following an iterative procedure [9]. Since the consensus algorithm requires

only the distributed communication between adjacent nodes [15], it can be used as the basis of a distributed control approach. The consensus algorithm is mainly used in the automatic control and the optimal scheduling in the domain of electrical power system [16]. There are two types of consensus algorithm: continuous consensus and discrete consensus. Considering the discrete control is predominantly applied in the actual system [17], the first-order discrete consensus algorithm is integrated into this work, as described by:

$$x_j[k+1] = \sum_{j=1}^N d_{ij} x_j[k], \quad i=1,2,\dots,N \quad (3)$$

where,  $k = 1,2,\dots,n$  is the discrete time index,  $x_i$  is the state variable of node  $i$ ,  $d_{ij}$  is the information weight factor between node  $i$  and  $j$ . The state variable,  $x_i$  can represent the physical quantities, e.g., voltage magnitude and output power.

Equation (3) is expressed in the following matrix form:

$$\mathbf{X}^{k+1} = \mathbf{D}\mathbf{X}^k \quad (4)$$

where  $\mathbf{X}$  is the system state variable vector and  $\mathbf{D}$  is the information weight matrix. If  $\mathbf{D}$  has all non-negative elements and the sums of each row are all ones, the consensus convergence of the system is ensured [17]. With a given consensus convergence accuracy  $\mathcal{E}$ , the system reaches the consensus convergence if state error of the  $k$ th interaction is less than  $\mathcal{E}$  [17], as expressed by:

$$\sum_{i=1}^N |x_i[k] - x_i[k-1]| < \mathcal{E} \quad (5)$$

To obtain a common equilibrium point among nodes, the information weight factors need to be properly designed [18].

### III. PROPOSED CONTROL SCHEME

The local control act as the lower level control to continuously monitor and mitigate overvoltage problems at the POC. Consensus-based distributed control is integrated as a higher level control to coordinate the droop controls. Fig. 1 illustrates the proposed control scheme in a LV MG.

#### A. Local Control as the Lower Level Control

Combination of  $Q$ - $V$  and  $P$ - $V$  droop control of PV inverters introduced in [5] is adopted in this work to solve the overvoltage problem. From Fig. 2, it is noticed that the  $Q$ - $V$  droop control is activated prior to the  $P$ - $V$  droop control (i.e.,  $V_{athQ} < V_{thP}$ ) with the aims of reducing the active power curtailment for overvoltage mitigation. Equation (1) reveals that  $P$ - $V$  droop control requires  $V_{thP}$  and  $m$ . According to the conventional  $P$ - $V$  droop control, the values of  $V_{thP}$  and  $m$  are constants. In this work, these droop parameters are adjusted by the consensus-based distributed control acting as the higher level control.

#### B. Consensus-based Distributed Control as the Higher Level Control

To restrict the dependence of an extensive communication network and realize the plug-and-play features of a MG, the consensus-based distributed control using a sparse communication network is integrated into this proposed control

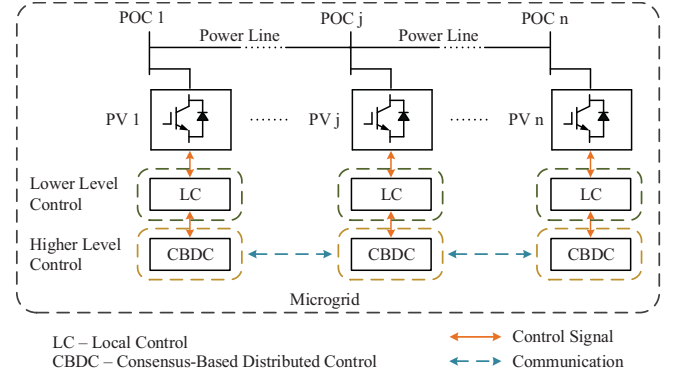


Figure 1. Proposed control scheme diagram of a LV MG.

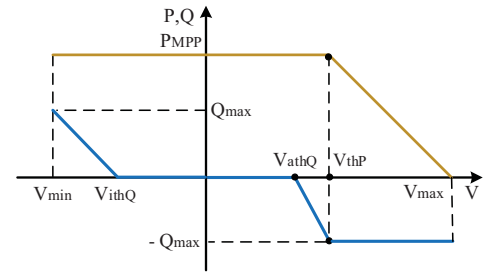


Figure 2. The droop control mechanism of a PV inverter.

scheme. Each of PV inverters acts as a controllable unit, which communicates and shares its state variables to neighboring units via sparse communication. The consensus-based distributed control aims at curtailing the active power to achieve the PV output active power,  $P_{net}$  proportional to  $P_{MPP}$  at given solar irradiation and ambient temperature. Let  $\beta$  denotes the injection ratio PV inverters, as expressed by:

$$\beta_i = \frac{P_{net}^i}{P_{MPP}^i} \quad (6)$$

The control objective is to determine  $\beta$  such that (10) is satisfied in the equilibrium point.

$$\beta_1 = \beta_2 = \dots = \beta_n \quad (7)$$

In this way, the required curtailment of active power to mitigate the overvoltage issue is equally shared among available inverters in proportion to the generation capacity. The first-order discrete consensus algorithm as described by (3) is utilized to obtain the desired  $\beta$ . As described, the information weight matrix,  $\mathbf{D}$  needs to be properly designed. The Metropolis method proposed in [19] is applied to construct  $\mathbf{D}$  and can be mathematically presented as follows:

$$d_{ij} = \begin{cases} \frac{1}{[\max(n_i, n_j)]} & j \in N_i \\ 1 - \sum_{j \in N_i} \frac{1}{[\max(n_i, n_j)]} & i = j \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

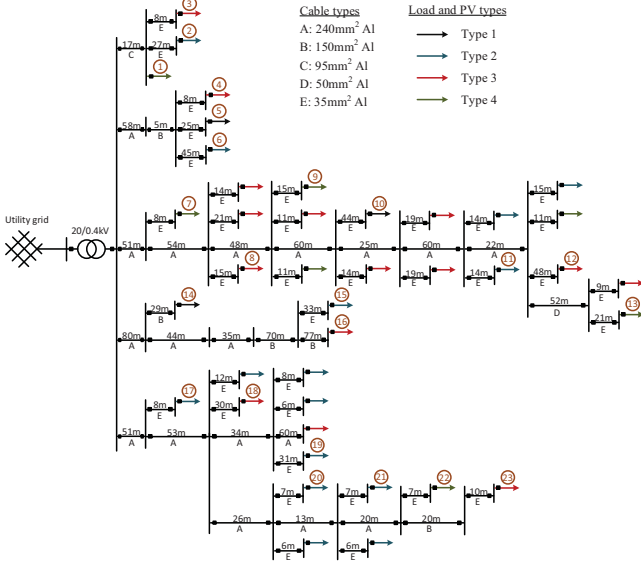


Figure 3. Simulation test network.

where  $n_i$  and  $n_j$  are the numbers of adjacent nodes to which node  $i$  and  $j$  connect respectively. The remaining part of the algorithm focuses on recalculating the  $P$ - $V$  droop parameters,  $V_{thP}$ , and  $m$ , of PV inverters with the injection ratio,  $\beta$  resulting from the consensus algorithm, as mathematically presented in (9) and (10) where  $P_{rate}$  is the rated active power of the PV inverter. These updated parameters are subsequently provided to the droop control as reference values.

$$V_{thP}^i = \frac{P_{rate}^i V_m^i - (1 - \beta^i) \cdot P_{MPP} \cdot V_{max}}{P_{rate}^i - (1 - \beta^i) \cdot P_{MPP}} \quad (9)$$

$$m^i = \frac{P_{rate}^i}{V_{max} - V_{thP}^i} \quad (10)$$

#### IV. TEST NETWORK AND DATA

##### A. LV Microgrid Network

A LV MG based on a French LV distribution network is used as the test network for the simulation as shown in Fig. 3. The test network consists of 39 households, is supplied from a 20/0.4 kV, 2 MVA MV/LV transformer and is comprised of underground power cables with high R/X ratio comparing to overhead lines.

##### B. Household Load Profile and PV System Data

All 39 households are equipped with uncontrollable loads, 23 households are installed with rooftop PV. In Fig. 3 the PV units are numbered inside the red circles. The household loads are obtained using average normalized load profiles of 400 residential customers [2]. The data of solar irradiation and ambient temperature are obtained from [2]. These data are used to establish four different kinds of household and PV capacities, which are illustrated in Fig. 3. Household load and PV system data are summarized in Table I.

##### C. Sparse Communication Network for Distributed Control

In this study, a sparse communication network is developed

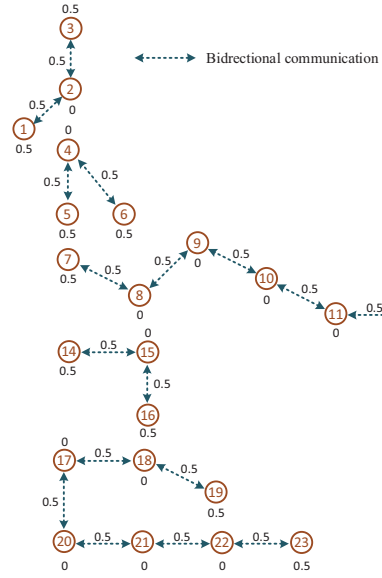


Figure 4. A sparse communication network for distributed control.

TABLE I. HOUSEHOLD LOAD AND PV DATA

| Load and PV                 | Type 1 | Type 2 | Type 3 | Type 4 |
|-----------------------------|--------|--------|--------|--------|
| Peak load [kWp]             | 2.37   | 3.63   | 2.70   | 2.53   |
| Load factor [-]             | 0.98   | 0.98   | 0.98   | 0.98   |
| Installed PV capacity [kWp] | 52.5   | 59.5   | 72     | 85.5   |
| Power factor [-]            | 0.9    | 0.9    | 0.9    | 0.9    |

at the level of the feeder. It implies that the communication between adjacent PV occurs only within the same feeder to which PV is connected. As such, the equal power curtailment for overvoltage mitigation is achieved at the feeder level. Fig. 4 illustrates the sparse communication network used in the simulation. In Fig. 4, the numbered red circles represent PV and the numbers along the dashed arrows denote the information weight factors defined by (8).

#### V. RESULTS AND DISCUSSION

Simulation has been performed in MATLAB/Simulink environment for two consecutive days. For the sake of comparison, four different control schemes are implemented.

- Scheme 1: No control
- Scheme 2: Conventional  $P$ - $V$  droop control with constant droop parameters
- Scheme 3: Consensus-based  $P$ - $V$  droop control
- Scheme 4: Combination of  $Q$ - $V$  droop and consensus-based  $P$ - $V$  droop control (Proposed control scheme)

For the sake of comparison between the control schemes,  $V_{thP}$  of 1.06 p.u. is set to the  $P$ - $V$  droop control. For Scheme 2,  $V_{thP}$  remains constant for all  $P$ - $V$  droop controls for two consecutive days. In contrast, for Scheme 3 and Scheme 4,  $V_{thP}$  is initially set to all  $P$ - $V$  droop controls, subsequently adjusted by the consensus-based distributed control to achieve the equal

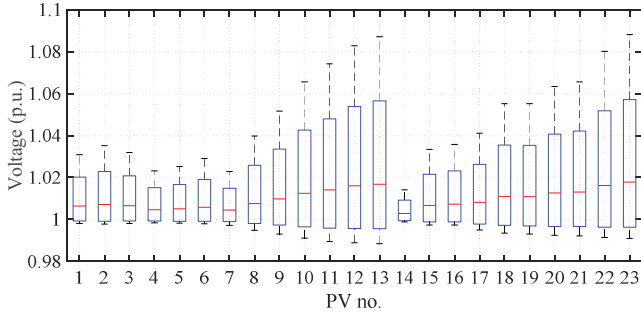


Figure 5. Voltage levels at the POC of PV.

contribution to overvoltage mitigation. Additionally, for Scheme 4,  $V_{athQ}$  of 1.05 p.u. is constantly set to the  $Q$ - $V$  droop control.

#### A. Overvoltage Mitigation

Fig. 5 illustrates the voltage levels measured at the POC of all PV in the test network without the application of any control. It is observed that the PV systems connected towards the end of a feeder experience higher voltage levels compared to the ones connected at the beginning of a feeder (PV nos. 13 and 23 compared to others). Feeder nos. 3 and 5 accommodating PV nos. 13 and 23 respectively are the longest feeders in the test network, thus having the largest voltage rise. In contrast, the other feeders have dramatically slower voltage rise.

Fig. 6 shows the voltage profiles, curtailed active power and absorbed reactive power of PV nos. 13 and 23 for four control schemes. It is observed that when voltage levels surpass  $V_{thP}$  of 1.06 p.u., these four control schemes yield different results of

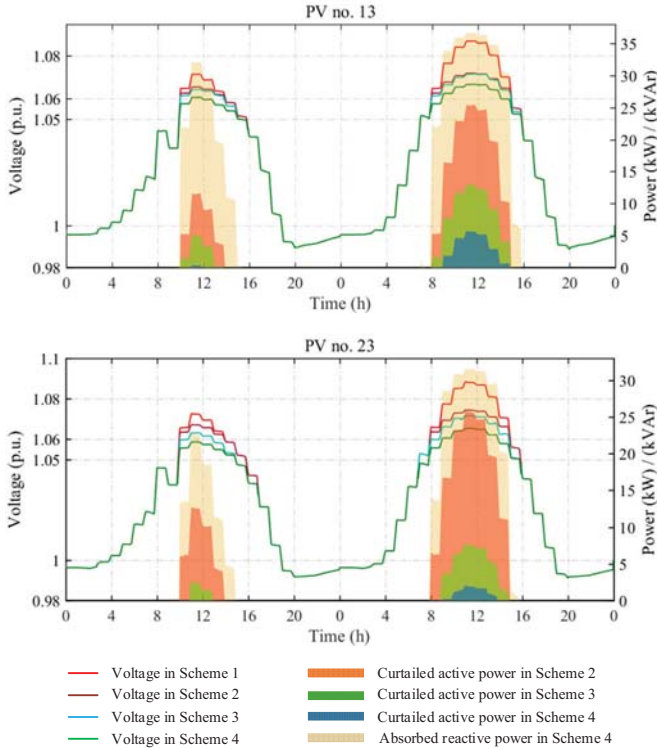


Figure 6. Voltage profiles curtailed active power and absorbed reactive power of PV nos. 13 and 23.

voltage profiles and output power of the PV inverters. Specifically, in Scheme 2, the conventional  $P$ - $V$  droop control limits the output active power of PV inverters in response to the voltage rise, resulting in a decrease in voltage levels. Compared to Scheme 2, in Scheme 3 the adoption of consensus-based distributed control provokes the lower voltage levels with a smaller amount of curtailed active power of PV inverters. Especially in Scheme 4 (the proposed control scheme), voltage levels and curtailed active power significantly reduce. This effective outcome is obtained because of the reactive power absorption according to  $Q$ - $V$  droop control. It proves that the activation of reactive power absorption prior to active power curtailment is capable of effectively declining the curtailed active power and mitigating the overvoltage problems.

Fig. 7 demonstrates the evolution of  $V_{thP}$  for  $P$ - $V$  droop control of PV systems connected to the feeder no. 3 and feeder no. 5 in Scheme 4 over the simulation time. It can be seen that  $V_{thP}$  of PV varies over the simulation time. When PV stops working during the night or no voltage rise exceeding 1.06 p.u. is detected,  $V_{thP}$  remains constant at 1.06 p.u. During the time period of voltage rise, which is greater than 1.06 p.u., the consensus-based distributed control continuously adjusts  $V_{thP}$ . Moreover, Fig. 7 shows that different PV inverters have different values of  $V_{thP}$ . Particularly, the PV systems connected to the beginning of the feeder have lower  $V_{thP}$  values while the PV systems connected to the end of the feeder have higher  $V_{thP}$  values. Recall that the voltage rises along the radial feeder. The adjustment of  $V_{thP}$  values increasing towards the end of the feeder is necessary to achieve equal power curtailment.

#### B. Fairness Evaluation

The amount of curtailed active power required to solve overvoltage problems in feeder no. 3 and feeder no. 5 is shown in Fig. 8 as the percentage of total PV energy generation. The

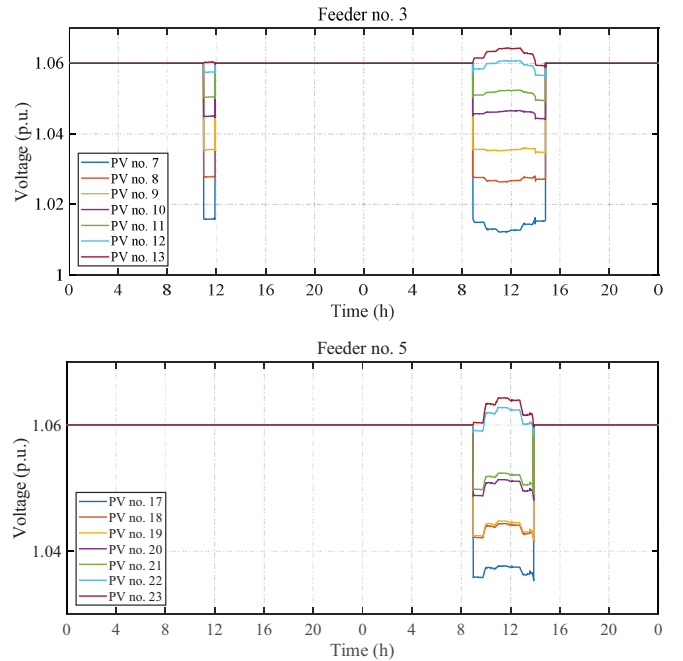


Figure 7. Voltage thresholds for  $P$ - $V$  droop control.

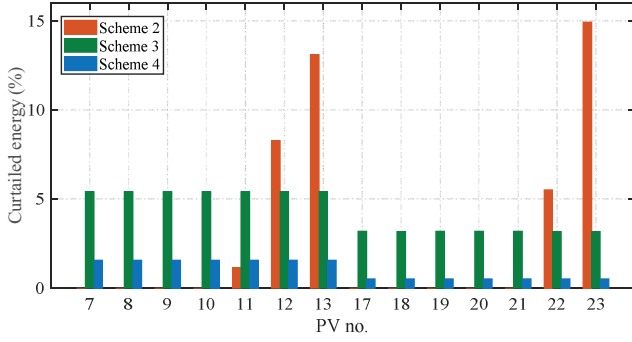


Figure 8. Percentage of curtailed energy of PV.

conventional  $P$ - $V$  droop control in Scheme 2 with a uniform value of  $V_{thP}$  leads to considerable variation in the amount of curtailed energy among the PV systems. In contrast, with the consensus-based  $P$ - $V$  droop control applied in Scheme 3 and 4, an equal energy curtailment is achieved in the feeder level. Especially, it is obvious in Scheme 4 that an insignificant amount of energy is curtailed because  $Q$ - $V$  droop control is implemented. This proves the effectiveness of the proposed control scheme in Scheme 4 to guarantee the fairness and minimization in active power curtailment of PV systems for overvoltage mitigation.

Key results including energy supplied, curtailed energy and maximum voltage in the test network are summarized in Table II. It is evident that Scheme 4 (the proposed control scheme) is the most effective approach to alleviate the overvoltage problems with lowest energy curtailment of PV systems.

TABLE II. SUMMARIZED MAIN RESULTS

| Properties             | Scheme 1 | Scheme 2 | Scheme 3 | Scheme 4 |
|------------------------|----------|----------|----------|----------|
| Maximum voltage (p.u.) | 1.088    | 1.074    | 1.072    | 1.067    |
| Energy supplied (MWh)  | 23.81    | 23.3     | 23.16    | 23.65    |
| Curtailed energy (MWh) | -        | 0.51     | 0.65     | 0.16     |
| Curtailed energy (%)   | -        | 2.13     | 2.73     | 0.68     |

## VI. CONCLUSIONS

This paper investigates a consensus-based distributed control for overvoltage mitigation in a LV microgrid with the aims of equal and minimum active power curtailment of PV systems. The lower level control is implemented by the local control at each of PV inverters utilizing  $Q$ - $V$  droop control, which is reinforced with additional  $P$ - $V$  droop control. The higher level control is performed using the consensus-based distributed algorithm along with a sparse communication network at the feeder level. Simulation results of the proposed control scheme are analyzed in comparison with different control schemes. The proposed method shows outstanding results for overvoltage mitigation while ensuring the fairness of curtailment. Additionally, the total curtailed energy has also been kept minimum. Hence, the proposed control can help DSO in solving the overvoltage problem and fairly treating the PV systems. It also helps PV owners in maximizing their revenue from selling the surplus energy.

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